

MURALI G

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Professional Summary

Embedded Systems Engineer with hands-on experience in developing embedded solutions using STM32, ESP32, and ARM Cortex-M microcontrollers. Experienced in Embedded C, FreeRTOS, hardware-software integration, peripheral driver development (ADC, SPI, I2C, UART, DMA, Timers), PCB bring-up, and system validation. Skilled in debugging, embedded Linux integration, and communication protocols, with practical experience in biomedical devices and IoT applications. Passionate about building reliable, high-performance embedded systems for healthcare, industrial automation, and robotics.

Education

Bachelor of Engineering, Biomedical Engineering (Hons.) 2022 – 2026
Bannari Amman Institute of Technology
CGPA - 8.2

Higher Secondary and Secondary Education 2019 – 2022
Nirmala Matriculation Higher Secondary School
HSC - 83.3% | SSLC - 96.4%

Work Experience

Embedded Hardware Intern 07/2025 – Present | Chennai
Qneuro India Pvt. Ltd.

- Developed and debugged embedded firmware on ARM Cortex-M microcontrollers for real-time EEG signal acquisition and biomedical data processing.
- Designed and implemented peripheral drivers using ADC, SPI, I2C, UART, DMA, GPIO, Timers, and interrupt-driven architectures.
- Developed an Automatic Gain Control (AGC) algorithm to dynamically calibrate IR and Red LED intensity based on skin tone, improving PPG signal quality.
- Implemented Bluetooth Low Energy (BLE) communication for reliable wireless transmission of biomedical sensor data.
- Worked on Linux-based Raspberry Pi systems for embedded integration, TCP/IP communication, and camera streaming applications.
- Assisted in PCB bring-up, hardware validation, circuit debugging, and embedded hardware integration.
- Conducted firmware testing, calibration, validation, and system verification using oscilloscopes, logic analyzers, and structured test procedures.
- Collaborated with cross-functional hardware and software teams throughout product development and validation.

Skills

Embedded Firmware & Systems

- Embedded C, Python, Linux
- FreeRTOS
- STM32 (ARM Cortex-M)
- ESP32 / ESP32-S3 (Xtensa & RISC-V)
- Interrupts, Registers, Debugging
- Testing and Quality Analysis

Communication Protocols

- UART, SPI, I2C, BLE
- Wi-Fi, TCP/IP
- MQTT, CAN, LoRaWAN

Embedded Hardware

- Hardware Integration
- PCB Bring-up & Debugging
- Schematic Understanding (Altium)
- Signal Validation (LTspice)

Tools

- STM32CubeIDE, STM32CubeMX
- Git, Visual Studio Code
- LTspice, Altium Designer
- Oscilloscope and Logic Analyzer

Projects

Deep Learning Cuff-less Blood Pressure Estimation - Engineered a continuous, cuff-less blood pressure monitoring system utilizing PPG and ECG signals. - Designed and trained a hybrid CNN-Bi-LSTM deep learning architecture to automatically extract spatial features and model physiological signal sequences. - Achieved high prediction accuracy for systolic and diastolic blood pressure, validating the model against synchronized ABP waveforms using MSE and correlation metrics.	2026
Predictive Machine Health and Control Hub - Developed a real-time sensor data acquisition project on ESP32 and STM32 platforms for early fault detection in industrial equipment. - Integrated cloud communication protocols for remote monitoring and predictive maintenance analytics. - Acquired and processed data from multiple sensors, including vibration, temperature, pressure, and current sensors, to monitor machine health. - Implemented UART-based communication between STM32 and ESP32, enabling reliable data transfer to a web-based dashboard for real-time visualization and analysis.	2025
Fetal Monitoring System - Developed an embedded project on ESP32 for motion-based fetal monitoring. - Interfaced sensors and implemented real-time data acquisition logic. - Integrated three IMU sensors to improve the accuracy and reliability of fetal movement detection during pregnancy. - Transmitted sensor data wirelessly to a web-based monitoring dashboard, enabling real-time visualization and tracking of fetal activity.	2024
Home Automation System - Developed firmware on ESP32-CAM for IoT-based home automation with sensors and actuators - Integrated peripherals and optimized energy usage using sensor-based control logic	2023

Achievements

Winner – MSME Idea Hackathon 5.0 Government of India Recognized and awarded ₹6 Lakhs for the project titled “Predictive Machine Health and Control Hub”.	2026
TN Skills Level - 2 Competition Naan Mudhalvan TNSDC Health and Social Care category	2024
Finalist - BITHack 23 Hardware edition for Home Automation System	2023

Certificates

Radio Frequency Applications Microchip Technology Inc.	PCB Design – Altium Designer Altium®
Introduction to IoT and Digital Transformation Cisco Networking Academy	Machine Learning Algorithms Great Learning

Languages

English Professional working proficiency	Tamil Native proficiency
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